

Designing Logic Pattern Templates for Counter-Argument Logical Structure Analysis

Anonymous ACL submission

Abstract

Counterarguments are a valuable tool in education for enhancing critical thinking skills. Despite their effectiveness, the logical attack structure of counterarguments in relation to their corresponding opponent argument remains unexplored due to its complexity. Towards tackling this challenge, in this work, we introduce “counter-argument (CA) logical structure analysis.”, a new, novel task. We first define 10 new CA logic patterns, each comprised of a unique template and slots. We then conduct an annotation study on top of 778 CAs using our patterns to create a new dataset. Our dataset achieves high annotator agreement (Krippendorff $\alpha=0.50$) and high coverage (86.5%). We perform preliminary experiments employing contemporary large language models to assess the feasibility of automating CA logical structure analysis. Our findings highlight the task’s inherent complexity within a straightforward framework, indicating exciting opportunities for further exploration. We publicly release our dataset and model scripts at <http://anonymous>.

1 Introduction

Counterarguments (CAs) are employed in education as an effective means to improve critical thinking skills, especially for interactive discussions such as debates (Roy and Macchiette, 2005; Liu and Stapleton, 2014; Johnson and Johnson, 1993; Nussbaum, 2008). Constructing an effective CA not only requires the ability to comprehend the logical structure of an opponent’s argument, but it also requires identifying where countering is most effective in order to construct a persuasive argument useful as an attack. Identifying vulnerabilities within an opponent’s argument is a fundamental aspect of critical thinking, as affirmed by numerous educational studies, such as the well-regarded Watson-Glaser Critical Thinking Appraisal (Watson and Glaser, 1952).

In the context of argumentation-based education, feedback provided by teachers plays a crucial role in enabling learners to enhance their skills effectively. Nonetheless, expecting all teachers to undertake this task is unrealistic, as it would require a significant amount of time to give tailored feedback to each student, and not all teachers undergo training in teaching argumentation (Driver et al., 2000). Developing a system that automatically gives feedback to students would be highly beneficial. Envisaging the downstream task of providing informative feedback to learners on the CAs they produce, this paper focuses on the design considerations for the task of CA logical structure analysis.

Various studies have explored the logical structure of arguments, with Argumentation Schemes (Walton et al., 2008) serving as a foundational framework for providing feedback on arguments (Macagno and Konstantinidou, 2013; Song and Ferretti, 2013; Song et al., 2014). Argumentation Schemes categorize approximately 60 argument forms commonly encountered in everyday discourse. Each scheme is paired with critical questions that challenge the validity of the argument, essentially offering feedback. For instance, Walton’s scheme of *Argument from Expert Opinion* is a type of argument that concludes that proposition *A* is true based on the endorsement of an expert *E*. This scheme prompts critical questions such as, “Is *E* an expert in the field that *A* is in?” and “Is *A* consistent with what other experts assert?”

The notion of schemes typology, as evident from Argumentation Schemes, proves advantageous in an educational context. Typology enables the generalization of individual cases, reducing the cognitive load and allowing students to apply past experiences to new cases (Rosch, 1978). However, despite its efficacy, there is no typology tailored for CAs. By establishing a typology for CAs, we can provide feedback for CAs in a manner akin to the mechanism of Argumentation Schemes.

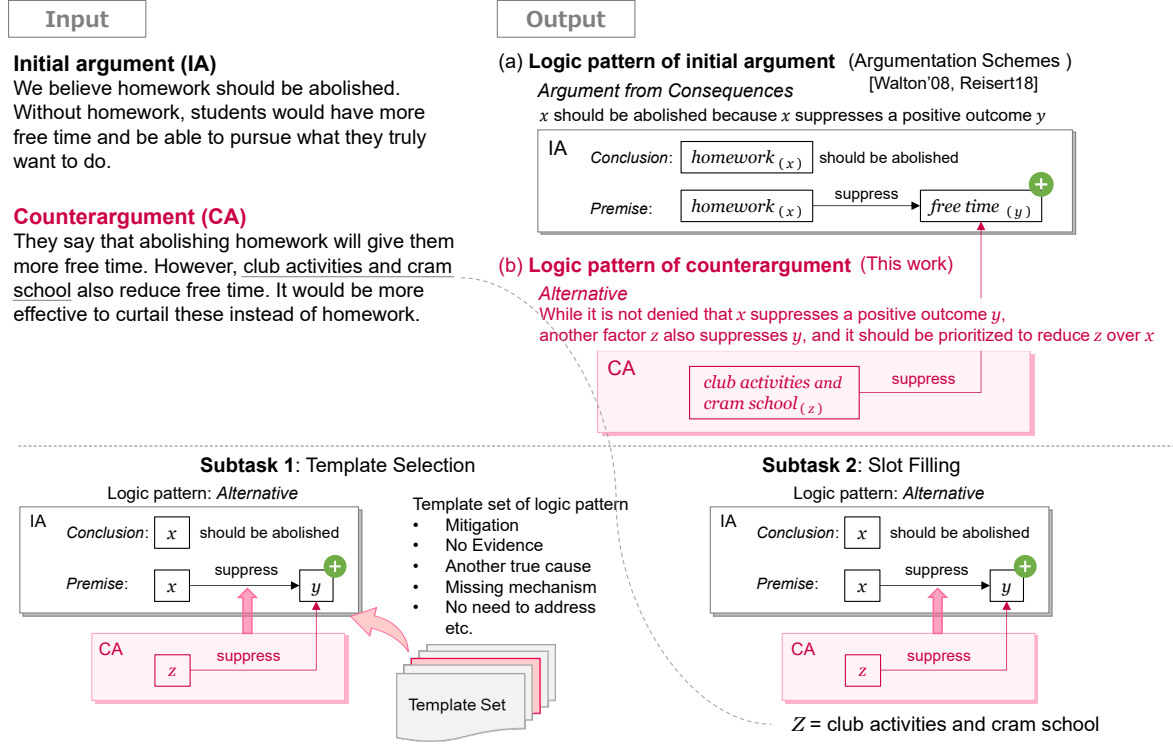


Figure 1: Our CA logical structure analysis task. Given a debate consisting of an initial argument (IA) and counterargument (CA), we i) select a pre-defined template to represent the CA's underlying logic in response to the IA and ii) fill in the template's slots (i.e., red text).

Figure 1 shows an example of an initial argument (henceforth, IA) and CA. In this specific example, the IA claims that the *homework* suppresses *free time*, and although the CA agrees with this logic, it provides an alternative to indicate that *homework* is not the main issue regarding *free time*, but *club activities and cram school* are the more significant issues that suppress *free time*. This pattern of suggesting alternative solutions (i.e., reducing “club activities and cram school” as an alternative solution to the “free time” issue) to these IAs emerges irrespective of the topic. Once these logical patterns are identified, it becomes possible to provide informative feedback to the CAs, such as “Why is the alternative proposed by the CA superior?”

In computational argumentation, several existing research studies have focused on CA analysis. Afantenos and Asher (2014) demonstrate the use of Segmented Discourse Representation Theory (SDRT) (Asher and Lascarides, 2003) to represent complex rhetorical moves, such as attacking the argument’s claim (i.e., the statement expressing the position of the arguer) while agreeing with a premise (i.e., supporting statement for the conclusion). LPAttack (Mim et al., 2022) provides an annotation scheme on kinds of attacks, focusing on

Argument from Consequences (Walton et al., 2008). However, no work has addressed the typology of logical patterns between a CA and its IA.

This work centers on CA logical structure analysis in a parliamentary debate setting. We first create a set of common logic patterns in CAs and construct a dataset on top of CAs. As the structure of logic patterns in a CA depends on the opponent’s argument and encompasses diverse argument types, we consider defining patterns for CAs using *Argument from Consequences*, a frequently occurring argumentation scheme (Feng and Hirst, 2011; Reisert et al., 2018; Visser et al., 2022).

Our work consists of two research questions: (i) How can we create an inventory of logic patterns in CAs that offers sufficient coverage and is feasible for annotation? (ii) To what extent can a language model identify these logic patterns of CAs? Our main contributions are as follows:

- We introduce CA logical structure analysis, a new task for representing the logical structure between a CA and its corresponding IA.
- We develop 10 new, distinct logic patterns for CAs based on the *Argument from Consequences* (Walton et al., 2008) scheme and

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annotate our patterns on top of an existing dataset of 778 debates (i.e., IA-CA pairs). Our dataset analysis reveals that our corpus has high annotator agreement (Krippendorff’s $\alpha=0.50$) and high coverage (86.5%).

- We evaluate the feasibility of our task by applying several language models with few-shot and fine-tuning on the constructed dataset.
- We publicly release our dataset and model scripts to facilitate further research in CA logical structure analysis.¹

2 Related work

2.1 Representation of attack relation

Several research studies exist on the argumentation structure, particularly focusing on the representation of attacks. In *abstract argumentation frameworks* (Dung, 1995), arguments are represented as a directed graph with nodes and edges as attack relations, allowing for the analysis of the acceptability of arguments based on some criteria. *ASPIC+* (Modgil and Prakken, 2014) is a known means to generate an abstract argumentation framework. *ASPIC+*, inspired by Pollock (1987)’s work, distinguish three kinds of abstract attack types: *rebuttal*, an attack on the conclusion of a defeasible inference, *undercut*, an attack on a defeasible inference step itself, and *undermine*, an attack on an ordinary premise. We go a step beyond these abstract types by understanding the more fine-grained reason for attack via our intricate logic templates.

In a study analyzing the discourse aspect of CAs, Afantenos and Asher (2014) employs segmented discourse representation theory (SDRT) to describe what is both acknowledged and denied within a CA. LPAttack (Mim et al., 2022) focuses on *Argument from Consequences* and proposes an annotation scheme for defining how causality and value are attacked. Although these works paved the way for exploring the representation of CA structure, they did not focus on the logical relation between a CA and its corresponding IA.

Argumentation Schemes (Walton et al., 2008) is a fundamental work that focuses on representing the logical structure of argumentation through schemes, where each scheme serves as a template for a type of argument commonly used. Each scheme primarily focuses on a micro-structure (i.e.,

premise and conclusion) within a single argument, opposed to the relationship between two opposing arguments. We expand upon the original notion of Argumentation Schemes by representing the structure of an CA argument in relation to its corresponding IA.

2.2 Attack identification task

Research has been conducted in Argumentation Mining (Mochales and Moens, 2011; Peldszus and Stede, 2013; Stede and Schneider, 2018; Lawrence and Reed, 2019) to automatically analyze argument structures. In a study targeting monological arguments, Stab and Gurevych (2014, 2017) developed the Persuasive Essay Corpus to identify both support and attack relations among argument components. Peldszus (2014); Peldszus and Stede (2015) created a dataset annotating rebuttals and undercut relations for online micro-texts. Similar to Argumentation Schemes, these studies focus on identifying relations (i.e., *attack*) between the argument components within a single argument.

Previous studies have focused on dialogical arguments by introducing a task to determine whether two arguments are in (dis)agreement, utilizing data from online discussion forums (Murakami and Raymond, 2010; Walker et al., 2012; Boltužić and Šnajder, 2014; Sridhar et al., 2015; Rosenthal and McKeown, 2015), peer reviews (Cheng et al., 2020), and debate speeches (Menini et al., 2018; Orbach et al., 2020). Other studies have introduced a task to identify the conflict relation at a more fine-grained level in argument discourse units (Ghosh et al., 2014; Visser et al., 2019; Hidey et al., 2017). Such studies assist towards understanding which argumentative discourse units are being attacked, but they do not consider the details as to how and why the attacks are made.

3 Design of Logic Pattern Templates

As discussed in §1, we create common logic patterns of CAs. Following Argumentation Schemes (Walton et al., 2008; Reisert et al., 2018), our patterns are represented by templates and their placeholders (henceforth, slots). By observing actual CAs, we manually derive an inventory of templates.

3.1 Requirements

For creating an inventory of templates for representing the logical structure of CA between its opposing IA, we establish the following key criteria:

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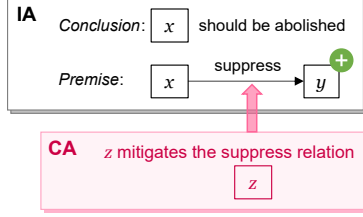
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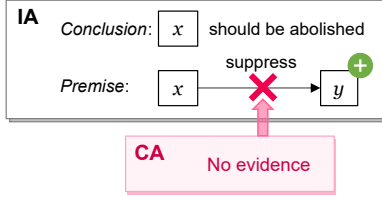
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Mitigation (Mig)

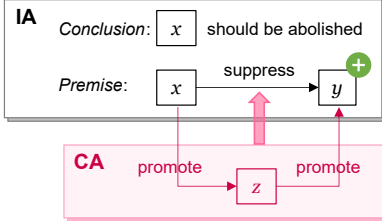
While it is not denied that x suppresses a positive outcome y , the causal relationship can be mitigated through the means of z .

**No evidence (No Evi)**

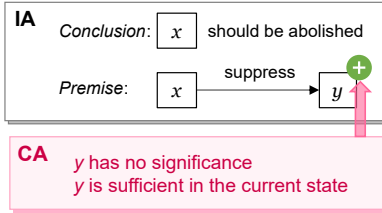
There is not sufficient evidence that x suppresses y .

**Missing mechanism #1 (MM1)**

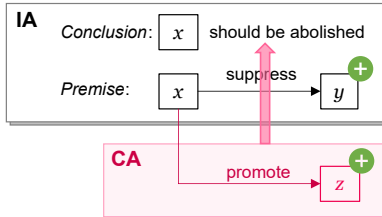
x promotes another factor z , and z promotes y . Therefore, x does not suppress y but rather promotes y .

**No need to address (NNA)**

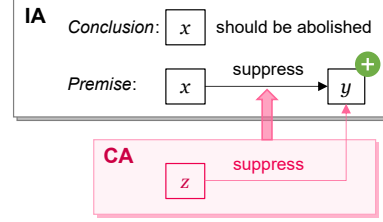
y is sufficient as it is, and there is no need to take action for y .

**Positive effects of a different perspective from y #1 (Dif Per1)**

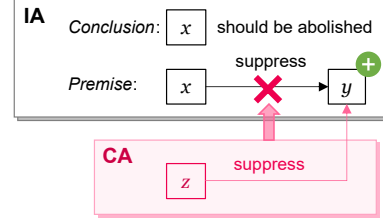
Since x promotes a positive outcome z , which is a different perspective from y , x should not be abolished.

**Alternative (Alt)**

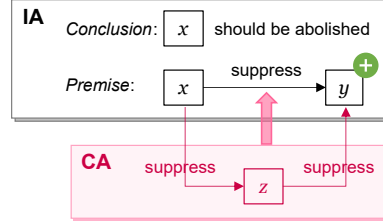
While it is not denied that x suppresses a positive outcome y , another factor z also suppresses y , and it should be prioritized to reduce z over x .

**Another true cause (ATC)**

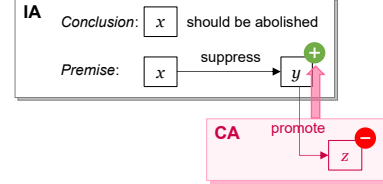
The true cause of suppressing y is z , not x ; therefore, abolishing x will not solve the problem.

**Missing mechanism #2 (MM2)**

x suppresses another factor z , and z suppresses y . Therefore, x does not suppress y but rather promotes y .

**Negative effect due to y (Neg eff)**

Since y leads to a negative outcome z , y is not a positive outcome.

**Positive effects of a different perspective from y #2 (Dif Per2)**

Since x suppresses a negative outcome z , which is a different perspective from y , x should not be abolished.

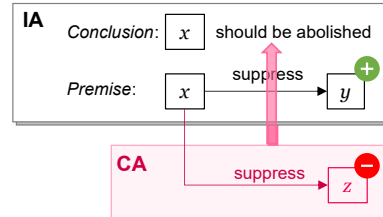


Table 1: Inventory of logic patterns we create for our new task of CA logical structure analysis. Our patterns were heavily inspired by *Argument from Consequences* scheme (Walton et al., 2008) and its template definition extended by Reisert et al. (2018). The templates represent how the CA attacks the logic of the IA, shown in red. The red arrows in the template diagram indicate which part of the IA (causality of premise, value of premise, conclusion) is being attacked.

First, we require that our logic templates are *comprehensive*, covering a wide-range of" CAs. This requirement is necessary to ensure that learners in a pedagogical setting receive appropriate feedback on a variety of different CAs.

Second, we require that our templates are *expressive* in that they should be capable of expressing the details of the logic sufficiently to provide informative feedback. This ensures that the feedback received by learners is specific and useful, especially in terms of critical thinking skill improvement.

Third, we require that the templates are *simple* and not overly complicated. This requirement is intended to simplify both human annotation and machine predictions.

Overall, creating a template set for all the requirements is a challenge. For the *comprehensive* requirement, a generic template (e.g., the argument x claimed by IA lacks persuasiveness) could be applied to several CAs. However, then our system could not satisfy the *expressive* requirement, as a generic template would not allow for useful feedback due to a lack of capturing important details such as logic. On the other hand, templates that express the finer details of the logic could provide informative feedback, thus fulfilling our *expressive* requirement. However, the more similar templates with minor differences are considered, the more complex the annotation becomes, thus failing to meet our *simple* requirement.

3.2 Inducing Templates

With the goal of meeting our requirements, we manually design our template set. For analysis in template design, we use TYPIC (Naito et al., 2022). TYPIC includes a total of 1,000 CAs and 10 IAs, with 100 CAs per each. For some CAs, feedback comments are also provided. TYPIC is ideal for our template creation, as it is suitable for deriving templates associated with feedback comments.

Observing TYPIC, we discover that the CA structure significantly varies depending on IA type. We, therefore, take the strategy of starting our exploration with a restricted range of IA types. Specifically, we utilize *Argument from Consequences* (Walton, 1999) for developing our templates. *Argument from Consequences* is a type of argument that concludes an action should or should not be brought about based on the good or bad consequences it may lead to. This scheme can be divided into Argument from Positive Consequences and Argument from Negative Consequences. The

latter is defined as follows:²

Premise: If action x is brought about, good consequences will plausibly occur.

Conclusion: Therefore, x should not be brought about.

We selected this argument type as our point of departure for the following two reasons. First, it is observed that this type of argument is most and highly common in discussions in which participants argue for or against a particular course of action (Walton, 1999; Feng and Hirst, 2011; Visser et al., 2019). Second, despite a limited range of argument types, it can still be highly beneficial in educational scenarios where students make counter-arguments to a given prompt IA because educators can control/choose the range of IA types to present as prompts.

To operationalize the Argument from Consequences scheme, allowing for a both annotation and model friendly, fine-grained logical representation, Reisert et al. (2018) created *argument templates*. Consider the following argument template for Argument from Negative Consequences:

Premise: x suppress y . AND pos(y).

Conclusion: neg(x).

In this template, x and y are *action* and *consequence* slots, respectively. neg(x) refers to the sentiment of x , and suppress refers to the relation between x and its triggering of consequence y . For creating our template set, we adopt similar ingredients, allowing for fine-grained representation of CA logic.

Through manual observation of CAs, we discover attacks can target the following: *premise causality*, *premise value*, and *conclusion*. Based on this finding, along with our requirement criteria, we create our inventory of 10 templates, as shown in Table 1, where each template represents how the CA attacks IA logic. These templates are designed to provide sufficient information for giving feedback. They explicitly detail how the attack is executed, enabling the formulation of useful critical questions as feedback for CAs.

Consider the "Positive effects of a different perspective from y #1". The CA attacks the IA's Argument from Negative Consequence-structured argument via the additional logic piece:

x promote z . AND pos(z).

²Please see Appendix A for an illustrated example.

Here, z is a placeholder for a consequence. This logic not only establishes an attack on the IA but also supports an implicit conclusion for Argument from Positive Consequences (i.e., x should be brought about.). Additionally, the following critical questions can be considered:

- Is there any direct attack on the IA stating that x suppresses y or that y is considered a positive outcome?
- Why is advantage of promoting z more important than disadvantage of suppressing y ?

3.3 Task Setting

As depicted in Figure 1, we formulate CA logical structure analysis as sentence-level sequential labeling task consisting of two subtasks: i) *template selection* and ii) *slot-filler extraction*.

Template Selection Given an IA and its CA, we first select a template from a predefined set of templates for each sentence. For our task, it is possible to express multiple templates within a CA by changing the template for each sentence. CAs often construct their arguments by combining multiple logic patterns. Therefore, we design the task to allow for segmentation by changing the patterns sentence by sentence. Since our task is sentence-level sequential labeling, its performance is automatically evaluated by sentence-level F1 score.

Slot-Filler Extraction The next step is to fill in the selected template slots. The template contains fillers (i.e., slot-fillers) that serve as a representation of key points for CAs. This task aims to extract phrases from the sentence that can be used as a template’s slot-filler within the CA. The performance is automatically evaluated using n-gram overlap.

4 Annotation Study

We describe the creation of our dataset using the TYPIC debate corpus and crowdsourcing.

4.1 Dataset

We utilize TYPIC (Naito et al., 2022), a dataset consisting of a debate topic, an IA, and a CA annotated with diagnostic comments. To be consistent with classrooms, where teachers use the same topics annually, TYPIC focused on *amicro-domain* approach, where a small set of topics with hundreds of counterarguments per each were considered. this allowed for the dataset to be used for

feedback construction while mitigating task difficulty. Such datasets are especially useful in an implicit discourse recognition task setting, a task that currently struggles with current LLMs (0.16 Macro F1) (Chan et al., 2024), where, for our task, we can cover a large range of typical CA logic patterns specific to a topic.

We additionally choose TYPIC for its reasonable size of IAs and paragraph-level CAs with multiple arguments, making the corpus appealing for our task of logic pattern analysis. In total, the corpus consists of two debate topics: *Ban death penalty (DP)* and *Ban homework (HW)*. We select 5 IAs and 490 CAs from the TYPIC corpus and extend upon it with a new topic: *Students should have a part-time job* with 3 new IAs and 288 new CAs, totaling 778 CAs (5,172 sentences).

4.2 Annotation Process

For each IA and CA pair, our annotation process is as follows. First, annotators carefully read both IA and CA. Next, annotators select all relevant templates per sentence in the CA, choosing “Not Applicable” if a template cannot be selected. Finally, annotators label the corresponding segments (i.e., consecutive sentences with same template label) for each chosen template, extracting phrase(s) from the CA that match the template slot-filler(s). (See Appendix B for actual annotation examples)

To ensure high-quality annotations, we incorporate both non-experts and experts. One characteristic of our task is that agreement occurs when the CAs are logically clear. Non-experts perform the annotations for such straightforward instances, while experts review the results for more challenging, disagreeing instances. We selected seven non-expert annotators via MTurk³.

Prior to main annotation, we conducted three rounds of calibration to address any disagreements. We interactively engaged in discussions with non-experts and updated our annotation guidelines accordingly. Main annotation was carried by 3 non-expert annotators per each of the 778 CA instances.

4.3 Annotation Results

To evaluate the feasibility of annotation, we measure inter-annotator agreement (IAA) and the coverage of the template set. The statistics of the dataset are shown in Table 3.

³<https://www.mturk.com/>

Quality of Annotation To measure annotation quality, we adopt aggregated labels using MACE (Hovy et al., 2013) with a threshold of 0.8 on top of our 778 CAs, resulting in 469 CAs with an aggregated gold label per each sentence.⁴ For these 469 instances, we calculate the IAA between three non-expert annotators and achieve a Krippendorff’s α (Krippendorff, 1980) of 0.50, indicating a high agreement. From this set, experts sampled 50 CAs and found 88.0% of labels were deemed valid. For the remaining 309 CAs, each sentence with a missing gold label was re-annotated by one expert. After the annotation, 50 CAs were verified by another expert annotator, determining that 78.5% of the re-annotated labels were valid.

We conduct a small, in-house study to observe the quality of annotator slot-fillers. Two expert annotators reviewed 43 slot-fillers across 18 patterns and 16 IA/CA pair instances, discussing disagreements afterwards to reach a consensus for disagreeing instances. After discussion, both agreed that 38/43 slot-fillers were correct. For disagreeing instances, annotators found that some instances required further context surrounding the slot-filler in the original text. As a result, only 2/43 slot-fillers were agreed to be incorrect.

Overall, by having non-experts annotate straightforward instances and experts review the complicated instances with low clarity, we were able to achieve high-quality annotations without significantly compromising scalability.

Coverage of Template Set We calculate the coverage of the template set and determine that 87.5% of sentences in our CAs contained a template besides “Not Applicable”. The majority of CAs can be represented within our template set.⁵

4.4 Analysis of Disagreement

We analyzed the types of disagreement to gain insight into our task. During our expert manual correction, we found that several CAs can consist of multiple reasoning patterns per sentence. This is the case, particularly for CAs that are not clearly written. For such CAs with multiple interpretations, we may want to provide feedback that they could be revised to reduce their ambiguity. Therefore, our task inherently allows for cases with multiple reasoning interpretations. In total, 134 CAs contained

⁴Experts reviewed instances in the bottom 20% of entropy for aggregated label probabilities calculated using MACE.

⁵Please see the Appendix C for the distribution of our templates

multiple reasonable interpretations for certain segments. Our dataset includes all labels for cases with multiple possible interpretations.

5 Model Experiments

Towards automatic CA logical structure analysis, we design our experiments to assess the performance of current large language models (LLMs) on our created dataset and to identify any challenges in automatically solving our task.

5.1 Automatic CA logical structure analysis

General Settings We employ fine-tuning and few-shot learning for training generative models to perform 1) *template selection* and 2) *slot-filler extraction* together in a general text2text manner. The format of model’s input and output is shown in Figure 2. We utilize LoRA and 4-bit quantization techniques to fine-tune large models due to the GPU resource limit. Regarding the dataset in use, we divide CA instances that have single-label annotations into both *train* and *dev* sets, where *train* consists of 516 CAs and *dev* consists of 128 CAs. For testing the models, we utilize all 134 CAs with multi-pattern annotations (henceforth, *test*). We train our models until overfitting and evaluate each checkpoint on *dev*. The checkpoint with the highest evaluation score was selected for *test*. Regarding few-shot learning, we randomly selected 20 CAs (20-shot) for the same IA as the target CA from *train* for in-context examples, for each CA in *test*. To ensure robust evaluation, all experiments are conducted 3 times. The fine-tuning experiments are conducted with 3 different seeds.

Evaluation *template selection* and *slot-filler extraction* are evaluated with sentence-level F1 score and precision score, respectively. We utilize RougeL to evaluate the lexical overlap between the generated slot and the label. We consider the generated slot correct if the RougeL score between it and the label is higher than a pre-defined threshold (0.5). See Appendix F for details.

Results As shown in Table 2, the scores are low for both *template selection* and *slot-filler extraction*, even when evaluating models’ generation with multi-pattern annotations (i.e. the generation is considered correct if it matches any one of the multi-pattern annotations). Specifically, the precision scores for *slot-filler extraction* are exceedingly low. This is partly attributed to our strict

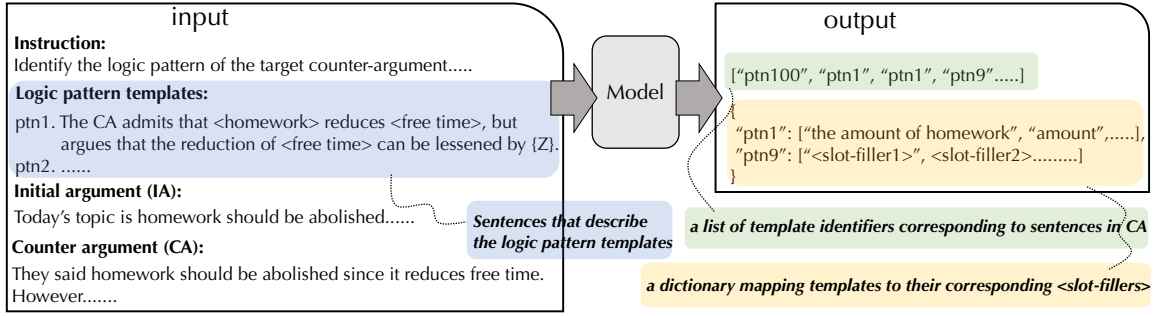


Figure 2: The input/output format of our model experiments.

models	F1 (patterns)	Precision (slots)
GPT-4	0.549	0.153
GPT-4-turbo	0.545	0.147
GPT-3.5-turbo	0.604	0.204
Mixtral8x7b	0.522	0.092
Yi-34b	0.564	0.095
Llama-2-13b	0.491	0.047

Table 2: Averaged F1 and precision scores over results of three runs for template selection (*patterns*) and slot-filler extraction (*slots*), respectively. The results above the line are for few-shot learning experiments, while those below the line are for fine-tuning experiments.

evaluation method, which solely rewards lexical overlaps without factoring in semantic information, as the slot-fillers are intended to be extracted directly from the original passage. Nevertheless, the results indicate that our proposed task is significantly challenging for current LLMs to address using a general text2text setting.

5.2 Error Analysis

In order to gain deeper insights into models’ behavior, we sampled 50 CA instances from the best checkpoint results of various models and conducted manual investigations on the generated outputs. Given the similar error trends we observed across different models, we will present an analysis of one model below. As mentioned in Section 4.4, a single CA may have multiple reasonable interpretations. We observed this phenomenon in the model’s output as well, where approximately 18% of the generated logic pattern templates from the sampled CA instances were reasonable but did not match the gold labels. Among those instances, we observed that models sometimes confused similar templates (e.g. MM1 and Dif Per1), a phenomenon also observed among human annotators. Additionally, we also observed that for approximately 11% of the instances, the generated templates matched the gold

labels, but the segmentation differed. Given the difficulty human annotators face in agreeing on a single segmentation (i.e., there are multiple possible segmentation for one CA), our task is highly challenging not only regarding the implicit logical reasoning required but also in terms of how to comprehensively and automatically evaluate the results. Moreover, regarding the *slot-filler extraction*, we found that out of all sampled 50 instances, 32% of them had slot-fillers that were too lengthy to include both the correct phrases and other information. Exploring methods for models to avoid generating redundant information would be a valuable direction for our future work.

6 Conclusion

In this work, we explored the new task of CA logical structure analysis. We first defined a new task setting and created a list of templates based on a popular Argumentation Scheme (Walton et al., 2008). We then conducted an annotation study to categorize 778 CAs with our templates and created a new dataset, achieving remarkable annotator agreement and coverage. Finally, we investigated the feasibility of utilizing LLMs to automatically identify templates and their corresponding slot-fillers. We observed that it is challenging to automate our CA logical structure analysis task with current LLMs utilizing a general text2text generation paradigm.

In our future work, we aim to create a feedback template set for our logic pattern templates and explore more advanced modeling methods to automate the CA logical structure analysis task. We also plan to extend our logic pattern template set to include other Argumentation Schemes as well. Finally, we plan to model methods for avoiding the generation of redundant information.

7 Limitations

Pattern creation For our task of CA logical structure analysis, we create our patterns based on the argumentation scheme *Argument from Consequences* (Walton et al., 2008) due to its frequency of usage in texts (Reisert et al., 2018). However, there are more than 60 argumentation schemes, so there is still room to explore other common argumentation scheme logic patterns.

Data Our work is limited to 3 topics for our experiments. Ideally, we would like to explore a wider range of topics. However, the cost of annotation prohibits us from currently expanding our corpus even further. Furthermore, while the progression of LLMs has improved performance in many tasks, implicit discourse recognition still remains a challenge. Chan et al. (2024) reported that in-context learning with LLMs achieved a Macro-F1 score of 0.16 for identifying implicit discourse relations in Penn Discourse Tree Bank. Opposed to collecting counterarguments on various topics in a broad and shallow manner, we aim to focus on a specific topic with a large amount of counterarguments. Such datasets collected with this approach are scarce and serve as valuable resources to assess the current limits of LLMs, as such datasets make it possible to cover typical CA logic patterns specific to each topic, which simplifies the problem by reducing cases that a model has never encountered before. Therefore, for the purpose of CA logical structure analysis, we require a dataset with a small number of topics with a large amount of CAs. Additionally, when considering actual usage scenarios, often only a few topics are immediately useful. For example, in a high school classroom, three topics may be sufficient. There is no need to prepare different topics for each class, nor is it necessary to change them every year.

Prompt Formatting for Model Experiments In this work, we experiment with only one type of input and output format for fine-tuning our model and few-shot learning. Thus, our work is limited in that there are more input and output formats that we can explore.

8 Ethical Statement

Crowdsourcing In this work, we conduct crowdsourcing experiments. For these experiments, the reward for annotating one CA was \$4.00. The work time per annotation was about 10 minutes,

and workers were paid more than the minimum wage. Additionally, bonuses were given for their hard work.

Acknowledgments

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A Argument from Consequences

The template for the IA used in this paper is shown in Figure 3.

Conclusion:

x should not be brought about.

Premise:

If x is brought about, then bad consequences y will occur.

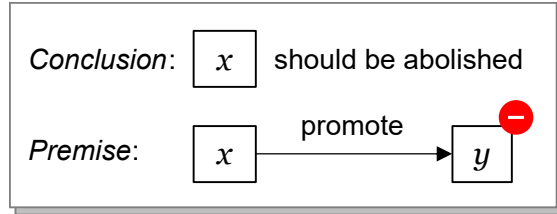


Figure 3: Argument from (Negative) Consequence scheme from Walton (Walton et al., 2008) which we utilize as a basis for our CA categorization task. The template defined by (Reisert et al., 2018) is used as a template for IA.

B Annotation Example

An example of the annotation is shown in Figure 6.

C Distribution of Templates

The distribution of templates is shown in Figure 4.

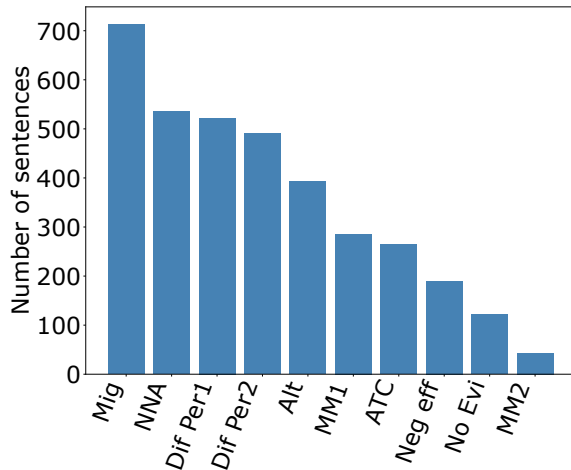


Figure 4: Template distribution of our dataset of 778 CAs, spanning 5,172 sentences. We show the distribution in terms of template per sentence.

D Disagreeing Annotation Example

Figure 5 shows an example of two disagreeing annotations which were labeled correct by our expert

annotator. Here, the logic can be interpreted as both “Missing Mechanism” and “Positive Effects of a Different Perspective from y ”. Interpretation 1 suggests that by promoting “their own initiative”, homework actually suppresses “passive in character”, contrary to the IA’s claim that it promotes passivity. Interpretation 2 suggests that homework offers other benefits, such as promoting “independence and self-discipline”, which are distinct from the IA’s perspective of promoting passivity. This disagreement arises depending on whether the IA’s “passive in character” and the CA’s “independence and self-discipline” are seen as opposing concepts (Interpretation 1) or as distinct concepts (Interpretation 2). These two patterns differ in the strength of their CAs (with Missing Mechanism being a stronger counterargument), and thus should be distinguished.

E Dataset Statistics

Table 3 shows the statistics of the dataset.

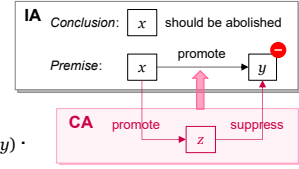
F Model’s Evaluation

For *template selection*, we evaluate the models’ generation with sentence-level F1 score. For *slot-filler extraction*, the generated slots are evaluated with precision score (i.e. the number of slots that are correct divided by the number of all generated slots). We first consider a slot incorrect when either 1) the predicted logic pattern template for which the slot is generated is incorrect since slots are dependent on logic pattern templates, or 2) the generated slot cannot be found in the counter-argument passage. If none of the above applies, we then determine whether a generated slot is correct by comparing the RougeL score between the target slot and each slot in the gold list. If the RougeL score is higher than a pre-defined threshold, we consider the target slot correct. For our experiments, we set the threshold to be 0.5.

IA: Today's topic is "Homework should be abolished". Our first point is that forcing students to do homework will make them passive in character... Therefore, for these reasons, homework should be abolished.

CA: ... However, in fact, compared to completing assignments in the classroom, homework is ideal for requiring students to work on their own initiative...By requiring students to work without the physical presence and real-time help of the teacher, homework promotes independence and self-discipline....

Interpretation 1: Missing Mechanism #1
homework (x), on the contrary, suppresses passive in character (y) because it promotes own initiative (z), which suppresses passive in character (y).



Interpretation 2: Positive effects of different perspective from y #1
homework (x), should not be abolished because it promotes independence and self-discipline (z), which are positive outcomes from a perspective different than passive in character (y).

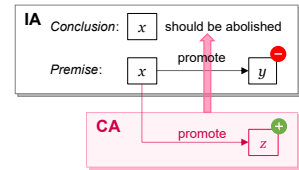


Figure 5: Examples of multiple reasonable interpretations possible.

IA ID	main point of IA	#sentences of CA
HW1	Abolishing homework gives students more free time	696
HW2	Homework makes students passive in character	690
HW3	Students memorizing the incorrect way to study with homework	749
DP1	Abolishing death penalty prevents misjudgement	622
DP2	Abolishing death penalty relieves executioners' stress	728
PJ1	Part-time job helps students learn responsibility and manners	572
PJ2	Part-time job helps improve academic performance	533
PJ3	Part-time job helps students recognize the importance of money	582
Total		5,172

Table 3: Statistics for our new corpus. We report the number of sentences for CAs associated with each IA.

IA

Today's topic is "whether homework should be abolished". We strongly believe that it should be abolished. We are going to talk about homework in every school in Japan. Our point is that [homework] (x) has harmful efforts on the educational development of school students. Some students have memorized the [incorrect way to study] (y) with homework. I'll give you some examples. In high school, students have lots of things to do, like hobbies, club activities, part-time jobs and so on. So how do students who have little time to do homework deal with it? Some of them do homework during class, which is inefficient not only with regard to the homework but also to the class. In a worst-case scenario, they copy it from their classmates. Of course, it means that they would learn nothing. But they receive a passing score as long as they finish homework without understanding how to properly study. Who thinks homework works adequately in these situations? We should abolish homework in order to let students understand the right and wrong ways to study. For all these reasons, we strongly believe homework should be abolished.

CA

They said that some homework is not efficient and effective, so homework should be abolished. [However, if homework is not efficient and effective, we should reform the homework by trying other ways.] [Furthermore, even if we abolish homework, it cannot necessarily make study more efficient and effective. In the first place, the method of homework is an effective way for all student to acquire enough skills by controlling proceeding. If we stop homework, the gap between those who can get enough skills and who cannot get is bigger than today education.] Therefore, abolishing homework is not allowed.

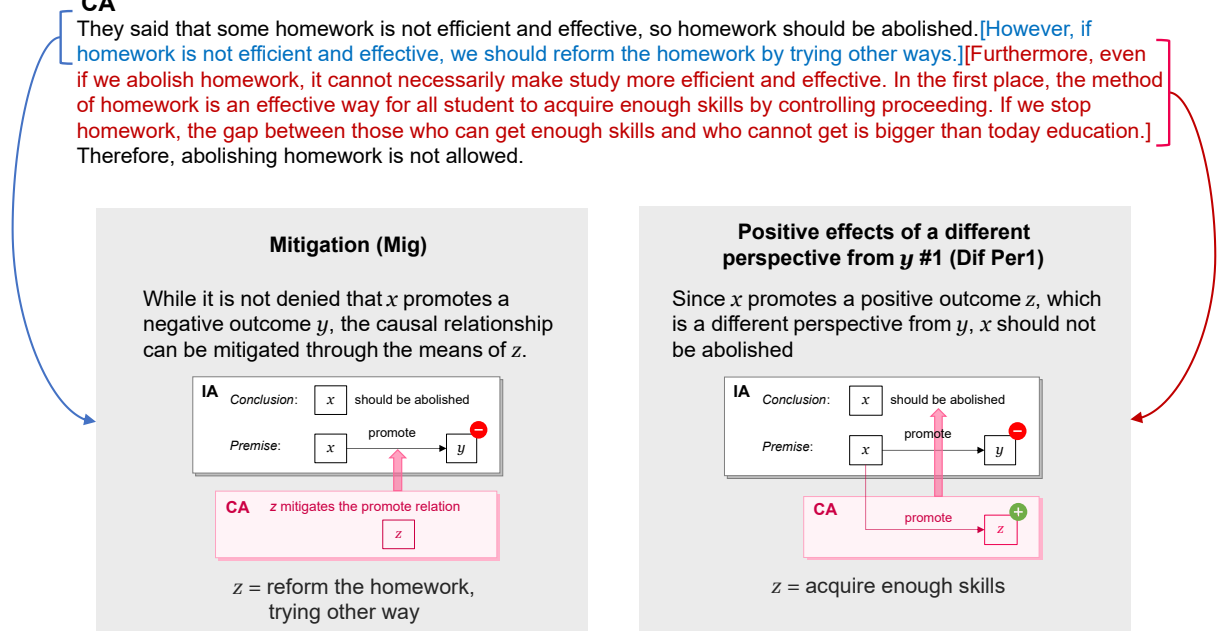


Figure 6: Annotation example of logic pattern templates and slot-fillers